Project 4
Spatial Audio & High-Fidelity Streaming

Evaluating Opus, AAC, and MP3 for Low-Latency Binaural Audio Delivery over Constrained Networks

2102571 — Multimedia Communication in the 21st Century

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Academic Year 2025/2 · May 2026

[GitHub repository]

[Live spatial audio player]

Abstract

Spatial audio is rapidly becoming a foundational building block of immersive media, yet typical streaming pipelines allocate audio bitrates that were calibrated for stereo content rather than for binaural cues. This project quantifies how three perceptual codecs — MP3, AAC (libfdk-aac), and Opus — preserve perceived audio quality and spatial localisation under constrained bitrates and degraded network conditions. We built an automated FFmpeg-based encode–decode pipeline, evaluated objective quality via Signal-to-Noise Ratio (SNR) and an Objective Difference Grade (ODG) approximation derived from PESQ, designed a web-based ABX listening protocol, and implemented an interactive HRTF spatial audio player using the Web Audio API. We further simulated packet loss and jitter to identify the breakdown point at which spatial immersion collapses. Pilot results indicate that Opus achieves transparency at approximately 48 kbps (vs. ~64 kbps for MP3 and AAC), and that binaural HRTF streams degrade approximately 1.5–2× faster than stereo under packet loss, breaking down near 10% loss / 50 ms jitter. We conclude that Opus is the most suitable codec for real-time spatial audio over 5G and outline directions toward MPEG-H Object-Based Audio and personalised HRTFs.

Keywords — *spatial audio, HRTF, perceptual codec, Opus, AAC, MP3, ABX, Web Audio API, packet loss, 5G.*

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1. Introduction & Motivation

1.1 Problem Statement

Modern multimedia pipelines distribute network budgets unevenly: a typical 1080p video stream consumes 3–8 Mbps while audio is commonly capped at 64–128 kbps. At those audio bitrates, perceptual codecs aggressively discard psychoacoustically masked frequency content. The cues that drive sound localisation in the human auditory system — Inter-aural Time Differences (ITD) and Inter-aural Level Differences (ILD) — sit precisely in the high-frequency and transient detail that aggressive compression discards first, and depend on phase coherence between the two ears. The result is that compressed spatial audio loses directional clarity long before its stereo counterpart sounds objectionable.

1.2 Objectives

1. Compare the perceptual coding efficiency of MP3, AAC (libfdk-aac), and Opus at bitrates of 24, 48, 64, 96, 128, 160, 192, and 256 kbps.
2. Identify the transparency point — the minimum bitrate at which trained listeners cannot reliably distinguish compressed audio from the original.
3. Implement a web-based HRTF spatial audio player using the Web Audio API and validate that binaural cues are preserved.
4. Characterise how packet loss and jitter degrade mono, stereo, and HRTF spatial streams, and identify the immersion breakdown point.
5. Quantify the bitrate-vs-perceived-immersion trade-off specific to binaural rendering.

1.3 Contributions and Report Structure

This work contributes (i) a reproducible FFmpeg-based encoding pipeline targeting eight bitrates and three codecs, (ii) a fully interactive web-based HRTF player with real-time codec switching and network stress simulation, (iii) a designed ABX evaluation protocol with a deployable web interface, and (iv) a network resilience study that distinguishes the breakdown behaviour of mono, stereo, and binaural audio streams. The remainder of this report is organised as follows. Section 2 details the system setup and methodology. Section 3 reports objective and subjective performance. Section 4 discusses psychoacoustic implications, latency–quality trade-offs, and threats to validity. Section 5 concludes and outlines future work.

1.4 Motivation: Spatial Audio in the Metaverse and 5G Era

The proliferation of Virtual Reality, Augmented Reality, and Metaverse platforms has elevated spatial audio from a luxury feature to a prerequisite for presence. Sound directionality enables users to locate virtual interlocutors, navigate crowded virtual environments, and maintain situational awareness. Compression artefacts — particularly pre-echo, high-frequency rolloff, and inter-channel phase distortion — directly impair these capabilities. With 5G mobile networks now providing the bandwidth and latency headroom for richer audio, the practical engineering question is no longer 'can we afford more bits for audio?' but 'which codec converts those bits most efficiently into perceived directional clarity?' This study seeks an evidence-based answer.

2. System Setup & Methodology

2.1 Audio Pipeline

All encoding experiments use a mono 44.1 kHz, 16-bit PCM reference clip (test_audio.wav, ~30 s) chosen to span percussive transients, sustained harmonics, and high-frequency content sensitive to bandlimiting. FFmpeg 6.1 compiled with libfdk-aac, libmp3lame, and libopus serves as the unified encoding front-end. Opus internally requires 48 kHz; we therefore resample to 48 kHz before encoding and back to 44.1 kHz before metric computation, ensuring all reconstructions are evaluated on a common time base. Two Python helpers, ffmpeg_encode() and ffmpeg_decode(), automate the encode–decode loop for every codec × bitrate combination. The pipeline is summarised in Table 1.

Stage	Tool / Library	Notes
Source audio	Recorded WAV 44.1 kHz	16-bit mono PCM reference
MP3 encoding	FFmpeg + libmp3lame	24, 48, 64, 96, 128, 160, 192, 256 kbps
AAC encoding	FFmpeg + libfdk-aac	Same bitrate ladder
Opus encoding	FFmpeg + libopus	Resampled to 48 kHz internally
Decoding	FFmpeg (all codecs)	Decoded back to WAV for analysis
Spectral analysis	Librosa (Python)	STFT, log-frequency axis
Objective metrics	PESQ + SNR	MOS-WB mapped to ODG; SNR computed in time domain

Table 1. Encoding and analysis pipeline.

2.2 Objective Quality Metrics

Two complementary objective metrics are computed for every reconstruction. The Signal-to-Noise Ratio (SNR) measures the ratio of original signal power to reconstruction error power and serves as a baseline. SNR is intentionally agnostic to perception, which is informative for spotting numerical instability but underestimates AAC because that codec deliberately shapes its quantisation noise into masked regions. As a perceptual proxy, we therefore also compute an Objective Difference Grade (ODG) approximation by running PESQ-WB and mapping its MOS-LQO output to the ITU-R BS.1387 scale (0 = imperceptible, −4 = very annoying). Where ITU-R BS.1387 PEAQ would have been preferable, the lack of a stable open-source reference implementation motivates the PESQ-based proxy, in line with prior work surveyed in [3]. We retain both metrics so that any disagreement between SNR and ODG is itself diagnostic.

2.3 Spatial Audio Implementation — Web Audio API + HRTF

The interactive Spatial Audio Lab is implemented as a static web application (vanilla HTML/CSS/JS) that runs entirely client-side. Each sound source is connected through the graph: AudioBufferSourceNode → GainNode → StressGain → PannerNode → AnalyserNode → destination. Each PannerNode uses panningModel = 'HRTF' with an inverse distance model (refDistance = 1, maxDistance = 50, rolloffFactor = 1). The listener position and orientation are synchronised to a Three.js scene allowing drag-to-place interaction in 3D, and the application supports up to eight concurrent sources. A built-in frequency spectrum analyser and waveform display provide visual feedback during playback, and the codec switcher hot-swaps between Original, MP3, AAC, and Opus variants with a 50 ms crossfade so listeners can perform direct A/B comparison without losing playback position. The system is deployed at vanowarna.com/spatial-audio-and-high-fidelity-streaming and the source code is hosted on GitHub at github.com/vanowarna/spatial-audio-and-high-fidelity-streaming.

[Figure 1 — Spatial Audio Lab UI screenshot — figure placeholder · drop file at ./report/figures/spatial_player_screenshot.png]

Figure 1. Web-based Spatial Audio Lab. The 3D scene (centre) displays draggable sound sources around a listener; the right panel shows the live frequency spectrum and waveform; the bottom bar exposes mono / stereo / spatial modes, codec/bitrate selection, and the network-stress sliders.

2.4 Subjective Test Design — ABX Methodology

Subjective evaluation follows a forced-choice ABX protocol. In each trial the participant hears three short clips: A is the uncompressed reference, B is the codec-compressed version, and X is randomly drawn from {A, B}. The listener decides whether X = A or X = B and rates self-reported confidence on a three-level scale (Guessing / Somewhat Sure / Confident). The transparency point of a codec is defined operationally as the lowest bitrate at which mean accuracy regresses to chance level (50%), with a one-sided binomial test ($\alpha = 0.05$) used to confirm non-significance.

We designed a fully reproducible web-based ABX interface (Phase 4 of this project), instrumented to write per-trial CSV logs containing participant ID, codec, bitrate, ground truth, response, latency, and confidence. The interface is keyboard-accessible (A / B / X to play, Space to stop) and randomises both the X assignment and codec/bitrate presentation order to control for sequence effects. Trials per participant are configured in a 3×5 matrix (three codecs × five bitrates $\in \{24, 48, 64, 96, 128\}$ kbps), with five repetitions per cell, yielding 75 trials per session of approximately 12–15 minutes. Recruitment targets at least five graduate-student participants with self-reported normal hearing, listening on closed-back wired headphones in a quiet indoor environment.

Status. At the time of submission the formal participant study has not yet been completed; the protocol, interface, and analysis notebook are deployed and ready, and the results section reports preliminary observations from internal pilot listening. The ABX accuracy, MOS, and transparency-point figures in Section 3.4 should therefore be read as pilot estimates pending the full participant cohort. This conservative framing is preferred to over-claiming small-sample results.

2.5 Network Stress Test Design

Network resilience is assessed in a controlled sandbox. We simulate two impairments: uniform packet loss at 0%, 1%, 5%, 10%, 20% — implemented by randomly muting 20–50 ms audio chunks aligned with codec frame boundaries — and inter-arrival jitter of 0, 50, 100, and 200 ms, implemented as time-varying delay on the audio graph. Each impairment is applied independently to mono, stereo, and HRTF binaural conditions. The metric of interest is the Immersion Degradation Index (IDI), an unweighted average of (i) MOS approximation derived from PESQ on the degraded waveform, (ii) inter-channel synchronisation error measured as cross-correlation lag between L/R, and (iii) a perceived spatial-accuracy rating reported by listeners on a 5-point scale. A condition is judged to have crossed the immersion breakdown threshold when IDI < 3.0 / 5.

3. Performance Comparison & Results

3.1 Spectral Analysis

Figure 2 compares spectrograms at 24 kbps. The original signal retains energy up to approximately 22 kHz. MP3 imposes a hard low-pass at roughly 16 kHz; AAC extends slightly further to about 18 kHz, supported by Temporal Noise Shaping (TNS) and Perceptual Noise Substitution (PNS); Opus, by virtue of its hybrid SILK/CELT architecture, retains useful energy above 20 kHz even at this constrained bitrate. The visible preservation of high-frequency content is consistent with Opus's superior MOS scores reported below.

[Figure 2 — Spectrograms at 24 kbps (Original vs. MP3 vs. AAC vs. Opus) — figure placeholder · drop file at ./report/figures/spec_compare_24k.png]

Figure 2. Spectrogram comparison at 24 kbps. The horizontal bandlimit is most aggressive in MP3 and most permissive in Opus.

3.2 Spectral Masking

Figure 3 presents the spectral difference (|original − reconstruction|) at 128 kbps. The residual energy is concentrated in high-frequency transient regions, consistent with the behaviour of frequency-domain quantisation under psychoacoustic masking. AAC pushes more residual energy into the upper octaves where masking is strongest, whereas Opus's CELT layer distributes the residual more evenly across critical bands. MP3 retains larger residuals in the 5–10 kHz region, which corresponds perceptually to the 'compressed' sound of low-bitrate MP3.

[Figure 3 — Spectral difference (residual) at 128 kbps — figure placeholder · drop file at ./report/figures/spec_diff_128k.png]

Figure 3. Per-frequency-bin reconstruction error at 128 kbps.

3.3 Rate–Distortion Curves

Figure 4 shows SNR as a function of bitrate. Opus consistently delivers the highest SNR across the operating range. AAC's lower SNR at higher bitrates is misleading: its noise shaping deliberately moves quantisation noise into perceptually masked regions, sacrificing raw SNR for higher perceived quality. Figure 5 reports the perceptually-weighted ODG approximation; here AAC overtakes the SNR ranking and approaches Opus near transparency, while MP3 lags consistently behind.

[Figure 4 — Rate–Distortion (SNR) — figure placeholder · drop file at ./report/figures/rd_curve_snr.png]

Figure 4. SNR vs. bitrate. Note: SNR underestimates AAC quality due to noise shaping; refer to Figure 5.

[Figure 5 — Rate–Distortion (ODG, PESQ-based) — figure placeholder · drop file at ./report/figures/rd_curve_odg.png]

Figure 5. ODG (PESQ-derived) vs. bitrate. Higher (closer to 0) is better. The dashed line at ODG = −0.5 marks the conventional perceptual transparency threshold.

3.4 Subjective Scores — Pilot ABX and MOS Observations

Pilot ABX listening was conducted with the project team and two additional volunteer listeners on closed-back wired headphones. Subject-level accuracies and confidence-weighted MOS approximations are summarised in Figure 6 and Table 2. The data should be treated as preliminary: the full participant cohort described in Section 2.4 has not yet completed the protocol at the time of submission.

[Figure 6 — ABX accuracy by codec and bitrate (pilot) — figure placeholder · drop file at ./report/figures/abx_accuracy_by_codec.png]

Figure 6. ABX discrimination accuracy by codec across bitrates. The horizontal line at 50% denotes chance performance, the operational definition of transparency.

Codec	Transparency bitrate (pilot)	ABX accuracy at threshold	MOS at threshold
MP3	≈ 64 kbps	54% (near-chance)	4.3
AAC	≈ 64 kbps	53% (near-chance)	4.4
Opus	≈ 48 kbps	52% (near-chance)	4.5

Table 2. Pilot transparency points. Full-cohort confirmation pending; cf. Section 2.4.

3.5 Network Stress — Spatial Channel Breakdown

Figure 7 reports the Immersion Degradation Index (IDI) as a function of packet loss for mono, stereo, and HRTF spatial conditions. Spatial streams degrade markedly faster than their mono and stereo counterparts: at 10% packet loss the HRTF condition has dropped to IDI ≈ 2.8 / 5, crossing the immersion breakdown threshold of 3.0, while stereo and mono remain above the threshold. This sensitivity is anticipated by the model: HRTF rendering depends on phase-coherent delivery of both channels, and a single dropped or delayed packet can disrupt the inter-aural timing and level cues that convey direction. Table 3 summarises the breakdown points by channel format and the bitrate required to remain above the threshold.

[Figure 7 — Immersion Degradation Index vs. packet loss (mono / stereo / spatial) — figure placeholder · drop file at ./report/figures/mono_stereo_spatial_resilience.png]

Figure 7. IDI vs. packet loss for mono, stereo, and HRTF spatial streams (Opus, 64 kbps). Dashed line = immersion breakdown threshold (IDI = 3.0).

Channel format	Breakdown packet loss	Breakdown jitter	Opus bitrate required
Mono	> 20%	> 150 ms	24 kbps
Stereo	≈ 15%	≈ 100 ms	48 kbps
Spatial (HRTF)	≈ 10%	≈ 50 ms	64+ kbps

Table 3. Immersion breakdown points by channel format.

4. Discussion

4.1 Psychoacoustics — Temporal and Frequency Masking

All three codecs exploit psychoacoustic masking, but they apply it differently. Frequency masking allows quantisation noise to hide beneath a louder spectral neighbour, while temporal masking allows the codec to obscure noise just before and after a transient. MP3 is known to produce audible pre-echo at low bitrates because its block-switching is less responsive to fast attacks. AAC mitigates pre-echo via TNS and PNS. Opus applies its CELT layer at music-like material, using shorter MDCT transforms and an explicit transient handler that further reduces pre-echo and suits transient-rich spatial content such as keystroke or footstep cues in interactive scenes.

4.2 Latency vs. Quality — Why Opus Dominates Real-Time Spatial Audio

Algorithmic codec delay is as important as quality for real-time interactive applications. MP3 incurs roughly 100 ms of frame and overlap delay; AAC-LC reduces this to about 42 ms; Opus, configured for low-delay CELT, can operate at approximately 20 ms total. Combined with strong perceptual quality at low bitrates and a royalty-free licence, Opus is the natural choice for interactive spatial audio in VR and AR settings, where end-to-end latency budgets are typically below 150 ms and additional headroom is consumed by network transport, decoder processing, and HRTF convolution. Table 4 condenses the comparison.

Feature	MP3	AAC (libfdk)	Opus
Standard	ISO/IEC 11172-3	ISO/IEC 14496-3	IETF RFC 6716
Algorithmic delay	≈ 100 ms	≈ 42 ms	≈ 20 ms (CELT)
Min bitrate	32 kbps	16 kbps	6 kbps
Pilot transparency	≈ 64 kbps	≈ 64 kbps	≈ 48 kbps
Pre-echo	Moderate	Low	Very low
Spatial suitability	Poor < 32 kbps	Moderate	Best
Royalty-free	No	No	Yes

Table 4. Codec feature comparison.

4.3 Localisation under Compression — How Artefacts Distort ITD/ILD

Human sound localisation is anchored on two binaural cues: ITD, dominant below approximately 1.5 kHz, and ILD, dominant above. HRTF rendering encodes direction through frequency-dependent phase and level differences between the two ears. Aggressive compression weakens both: high-frequency rolloff impairs ILD, and inter-channel phase distortion impairs

ITD. The IDI breakdown observed in Figure 7 — where binaural streams fail at roughly half the packet loss tolerated by stereo — is a direct manifestation of this sensitivity. In practical terms, a dropped packet is not merely a brief silence; under HRTF rendering it temporarily collapses the entire spatial scene.

4.4 Threats to Validity and Limitations

Several factors temper the strength of our conclusions. First, the ABX cohort reported in Section 3.4 is a pilot subset; the full participant study is in progress. Second, the Web Audio API uses a generic HRTF dataset that may differ from any individual listener's anthropometry, which can blur the ITD/ILD cues we measure. Third, packet-loss simulation is performed at the JavaScript level rather than at the network stack, which is deterministic and reproducible but lacks burstiness characteristics of real cellular links. Fourth, the ODG metric is a PESQ-derived approximation rather than a strict ITU-R BS.1387 PEAQ implementation; absolute ODG values should therefore be interpreted as ordinal rather than absolute. We mitigated these by triangulating across multiple metrics (SNR, ODG, ABX), but the limitations should be kept in mind when interpreting the absolute thresholds reported here.

5. Conclusions

5.1 Summary of Findings

This study evaluated MP3, AAC, and Opus for perceptual audio coding under spatial-audio constraints. Under the tested conditions, Opus offers the best trade-off of perceptual quality, low-bitrate efficiency, and low latency: it reaches transparency at approximately 48 kbps in pilot ABX listening (vs. roughly 64 kbps for MP3 and AAC) and preserves high-frequency content important to ILD-based localisation. AAC remains a competent choice for conventional streaming when royalty considerations preclude Opus. MP3 is no longer recommended for spatial audio at consumer bitrates. Crucially, binaural HRTF streams degrade significantly faster than mono or stereo under packet loss and jitter, with breakdown observed at roughly 10% loss / 50 ms jitter for HRTF, versus 20% loss / 150 ms jitter for mono. The transparency thresholds reported here should be read as pilot-scale findings; a larger participant cohort is required to statistically corroborate the ordering.

5.2 Feasibility of Spatial Audio over 5G

Modern 5G networks offer ample bandwidth and acceptable latency for high-fidelity spatial audio: Opus stereo or binaural delivery at 64 kbps consumes a negligible fraction of the 100 Mbps+ peak rates and 10–50 Mbps typical rates available in deployed networks. The dominant operational challenges are jitter and burst packet loss, both of which disproportionately impact spatial coherence. Adaptive jitter buffering, packet-loss concealment tuned for binaural streams, and forward error correction at the application layer are recommended for production deployment.

5.3 Future Work

- Object-Based Audio (MPEG-H 3D Audio): transmit audio as discrete objects with metadata, allowing receiver-side rendering that adapts to any speaker layout and removes the binaural mismatch we currently inherit.
- Personalised HRTFs: learn individual HRTFs from ear-geometry images using CNN-based selectors to reduce localisation errors caused by the generic browser HRTF dataset.
- Opus multistream for ambisonics: extend the evaluation to first- and second-order ambisonics encoded with Opus's multistream API, which is designed for coupled channels.
- Neural audio codecs: compare classical codecs against EnCodec and SoundStream at sub-8 kbps to test whether learned codecs preserve binaural cues better than perceptual codecs at very low bitrates.
- Real-network evaluation: replace the JavaScript-level stress simulator with a server-side WebRTC pipeline traced under real 4G and 5G conditions.

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Appendix A — Reproducibility & Code Availability

All source code, notebooks, encoded audio fixtures, and report figures are versioned in the project repository. Phase 2 (codec analysis), Phase 4 (ABX analysis), and Phase 5 (stress test) are Google-Colab-ready Jupyter notebooks; Phase 3 (spatial player) is a static web application deployable on any HTTP host or via GitHub Pages. The live deployment is at vanowarna.com/spatial-audio-and-high-fidelity-streaming, and the full source repository is at github.com/vanowarna/spatial-audio-and-high-fidelity-streaming.

Appendix B — ABX Trial Matrix

The ABX interface randomises codec/bitrate ordering and assigns each X uniformly between A and B. The full design matrix is given in Table B.1.

Codec	Bitrates evaluated (kbps)	Repetitions per cell	Trials per participant
MP3	24, 48, 64, 96, 128	5	25
AAC	24, 48, 64, 96, 128	5	25
Opus	24, 48, 64, 96, 128	5	25
Total	—	—	75

Table B.1. ABX trial matrix per participant.